

Statistical Data Analysis Using Python

Student Performance Dataset

Tools: Python, Pandas, NumPy, SciPy, Matplotlib

1. Introduction

This project performs statistical analysis on the Student Performance in Exams dataset. The analysis examines student academic performance using numerical and categorical variables. Python libraries are used for data loading, cleaning, descriptive statistics, statistical testing, and visualization.

2. Dataset Description

The dataset contains 1,000 student records and 8 original variables: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score, and writing score.

3. Data Preparation

- Loaded the CSV dataset using Pandas.
- Checked the dataset shape, column names, data types, missing values, and duplicate records.
- Removed duplicate rows where applicable.
- Converted Math, Reading, and Writing scores to numeric data types.
- Created an Overall Score variable using the mean of the three subject scores.

4. Descriptive Statistical Analysis

Subject	Mean	Median	Std. Dev.	Minimum	Maximum
Math	66.09	66	15.16	0	100
Reading	69.17	70	14.60	17	100
Writing	68.05	69	15.20	10	100

5. Categorical Analysis

Gender distribution contains 518 female and 482 male students. For the test preparation course, 358 students completed the course and 642 did not.

The analysis also compares average academic scores across gender, lunch type, test preparation status, and parental education groups.

6. Correlation Analysis

Pearson correlation analysis is used to measure relationships between subject scores. The strongest relationship is between Reading and Writing scores (approximately 0.95), indicating a strong positive association. Math also shows strong positive relationships with Reading and Writing scores.

7. Statistical Tests

- Independent-samples t-test: compares Math scores between male and female students.

- Independent-samples t-test: compares Math scores between students who completed test preparation and those who did not.
- One-way ANOVA: compares Math scores across parental education groups.
- Pearson correlation test: evaluates the relationship between numerical subject scores.
- A significance level of $\alpha = 0.05$ is used for hypothesis testing.

8. Visualizations

- Math score distribution
- Reading score distribution
- Writing score distribution
- Average scores by gender
- Average scores by test preparation course
- Score correlation heatmap
- Student scores box plot
- Average scores by parental education
- Average scores by lunch type
- Gender distribution
- Test preparation distribution
- Overall score distribution
- Subject-wise average scores

9. Key Findings

- Reading has the highest average score among the three subjects.
- Math has the lowest average score among the three subjects.
- Reading and Writing scores have a very strong positive correlation.
- Students' academic performance can be compared meaningfully across test preparation and demographic categories.
- Descriptive statistics and inferential tests together provide a more complete understanding of the dataset.

10. Conclusion

The analysis demonstrates a complete statistical data-analysis workflow using Python. The project covers data loading, cleaning, descriptive statistics, categorical analysis, correlation analysis, hypothesis testing, and visualization. The results can be used to understand patterns in student performance and demonstrate practical use of Pandas, NumPy, SciPy, and Matplotlib.

11. Project Output Structure

The notebook is designed to save visualization files in the charts folder and statistical result files in the reports folder. The cleaned dataset is saved separately for further analysis.

Prepared as part of Task 5: Statistical Data Analysis Using Python Tools.